Graphene and Its Applications

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Graphene

Graphene is a two-dimensional Material with one atom thick carbon arranged in a hexagonal cubic lattice structure.

Graphene is the most Conductive, thinner, strongest and flexible material in the world.

Found in graphite, coal.

Hard to manufacture in large quantities, R&D still limited within universities.

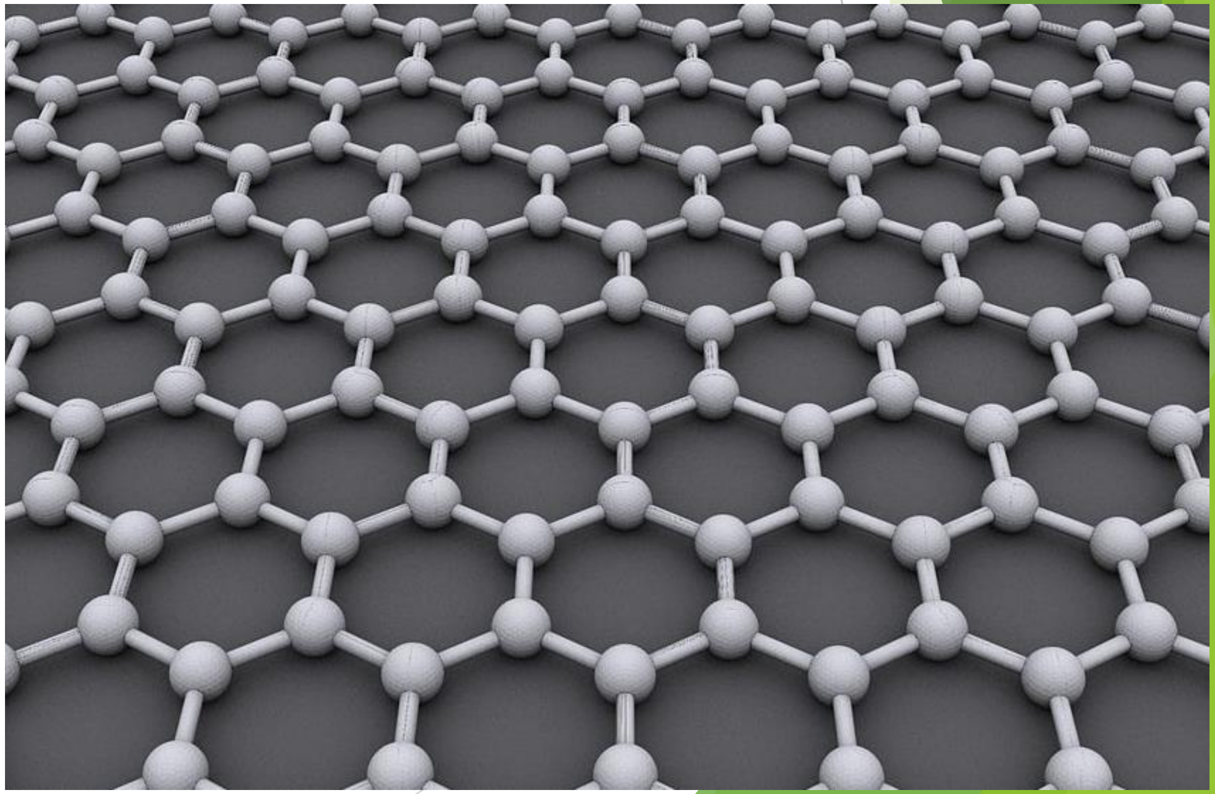


Fig: Wikipedia

Properties of Graphene
Some Comparison

The strongest, lightest, and most conductive material known to man

Bond length is .142 nm long = very strong bond

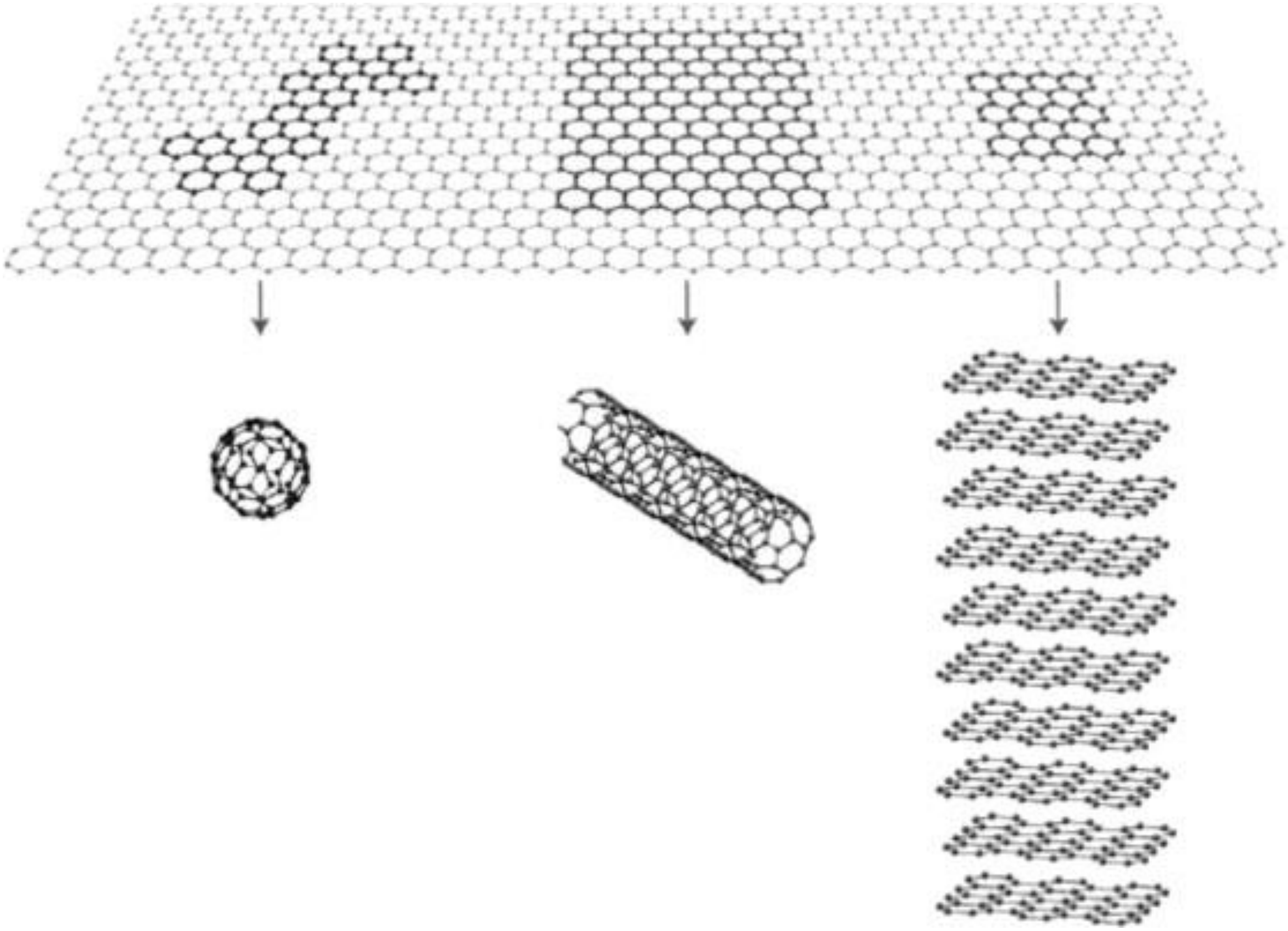
Ultimate tensile strength of 130 Gpa compared to 400 Mpa for structural steel means 150 times stronger.

Very light at 0.77 milligrams per square meter, paper is 1000 times heavier.

Single sheet of graphene can cover a whole football field while weighing under 1 gram.

Experimental graphene's electron mobility is 10,000 cm2/(V*s) and theoretically potential limits of 200,000 cm2/(V*s).

Schematic of Graphene



0D- Fullerene

1D-Nanotube (CNT)

2D- Graphene

3D-Graphite

Article, P. The rise and rise of graphene. *Nat. Nanotechnol.* **5**, 755 (2010).

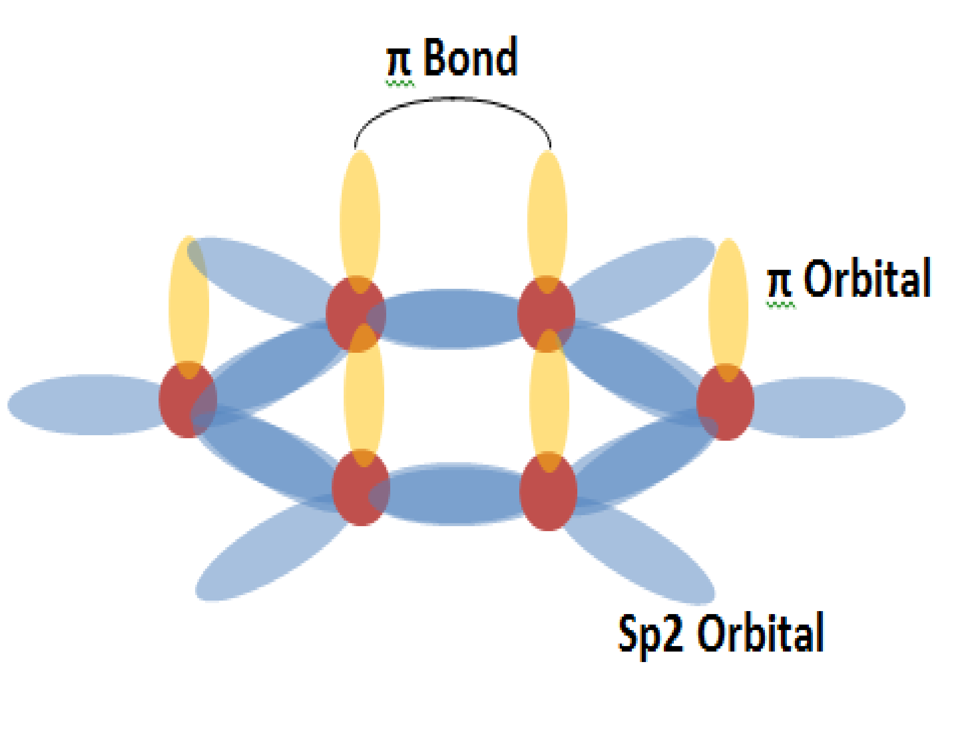
Properties of Graphene

Can anyone tell me why graphene is conductive? It is because of the SP2 hybridization that make the Pi-electron to move around and made it conductive.

Experimental graphene's electron mobility is 10,000 cm2/(V*s) and theoretically potential limits of 200,000 cm2/(V*s).

Table I Values of electron mobility and energy bandgap for Hall effect sensor materials

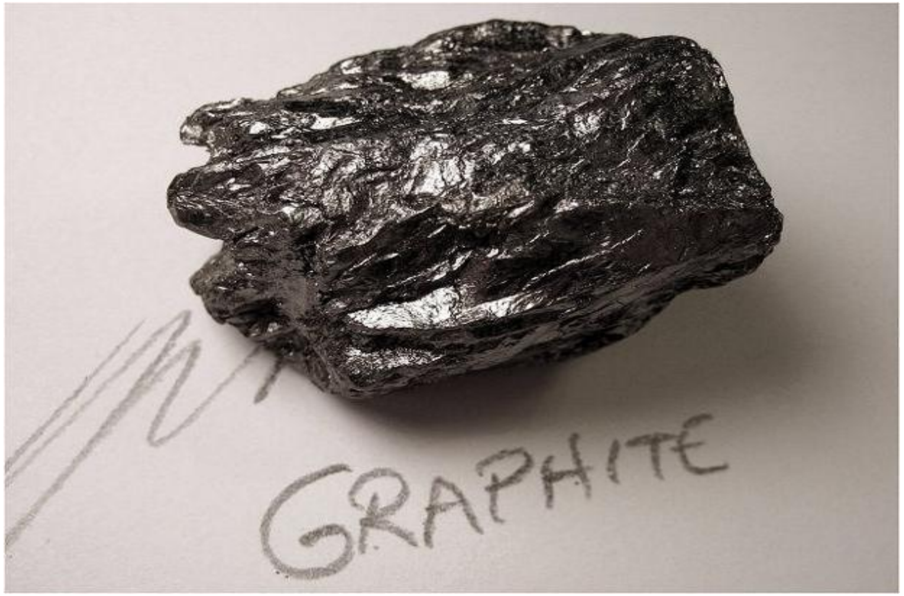
	$\mu(\mathrm{cm^2\ V^{-1}\ s^{-1}})$	E_g (eV)
Silicon	1,900	1.12
GaAs	8,800	1.43
InSb	78,000	0.17
InAs	33,000	0.35
InP	4,800-6,800	1.29
2-DEG	6,500-13,000	0.6-1.8



Discover of Graphene

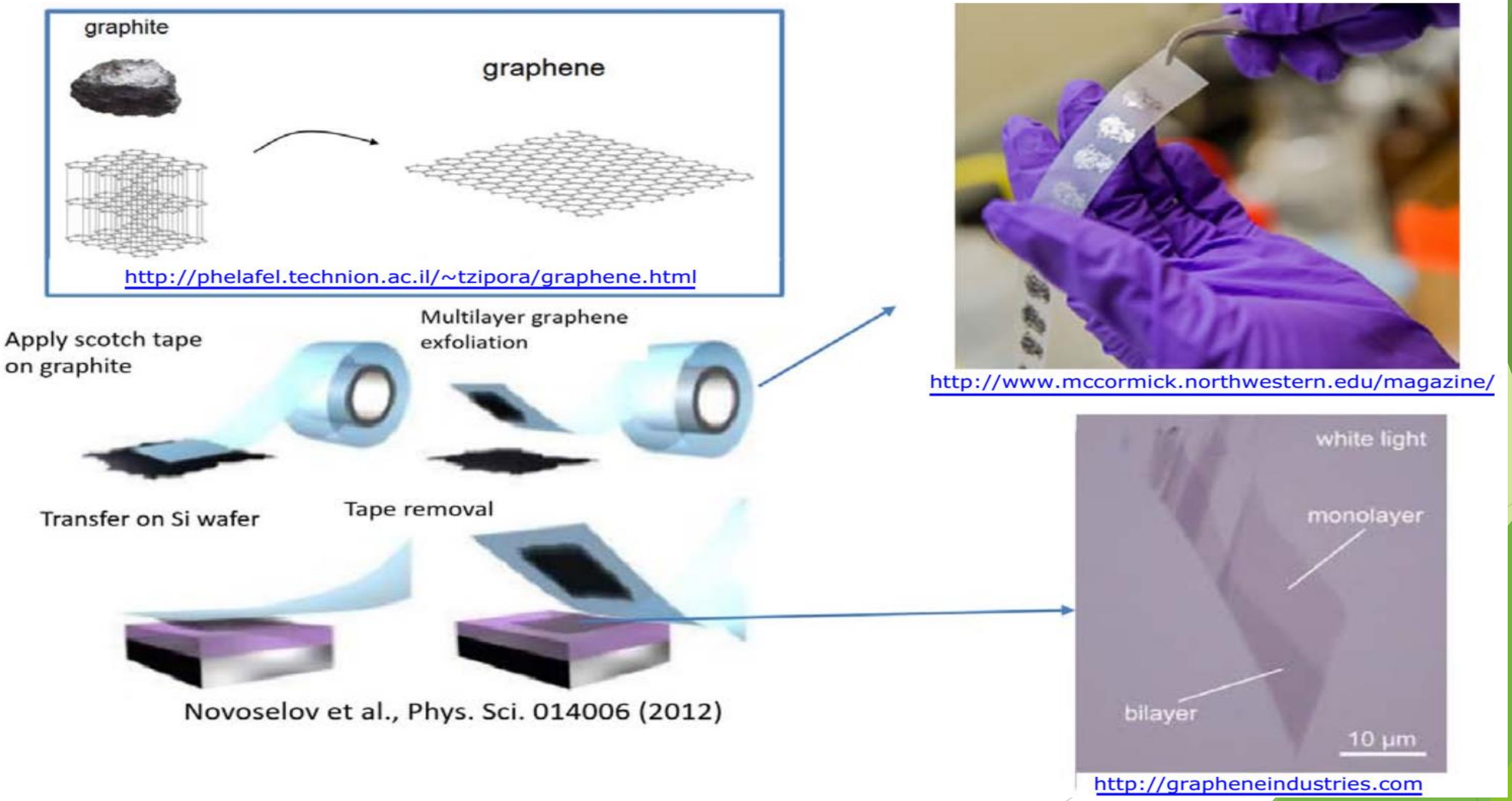
Studies on graphite layers goes back for past hundred year.

Andre Geim & Kontantin Novoselov from University of Manchester discovered graphene from graphite in 2004 by a process knows as the Scotch tape technique (physical exfoliation) and won Nobel Prize in Physics in 2010.



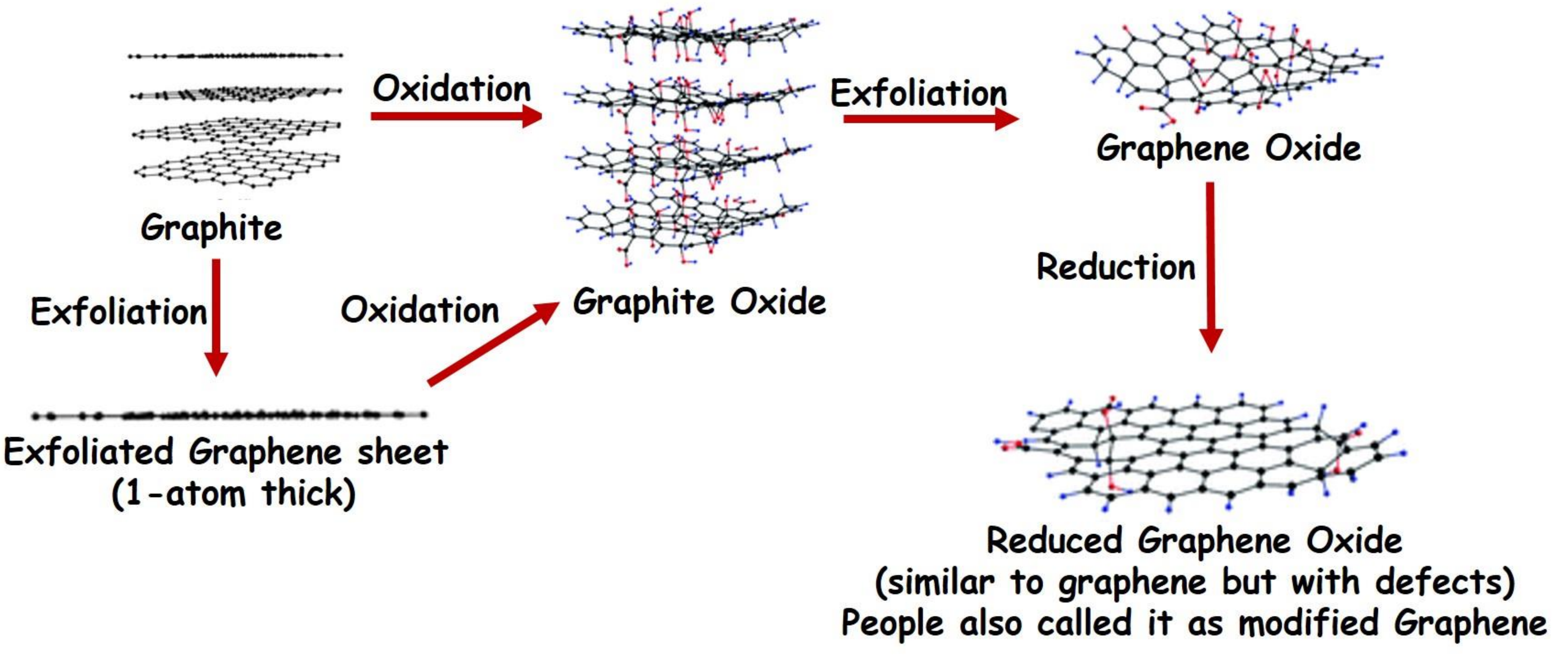
Graphene Synthesis

Mechanical Exfoliation



Graphene Synthesis

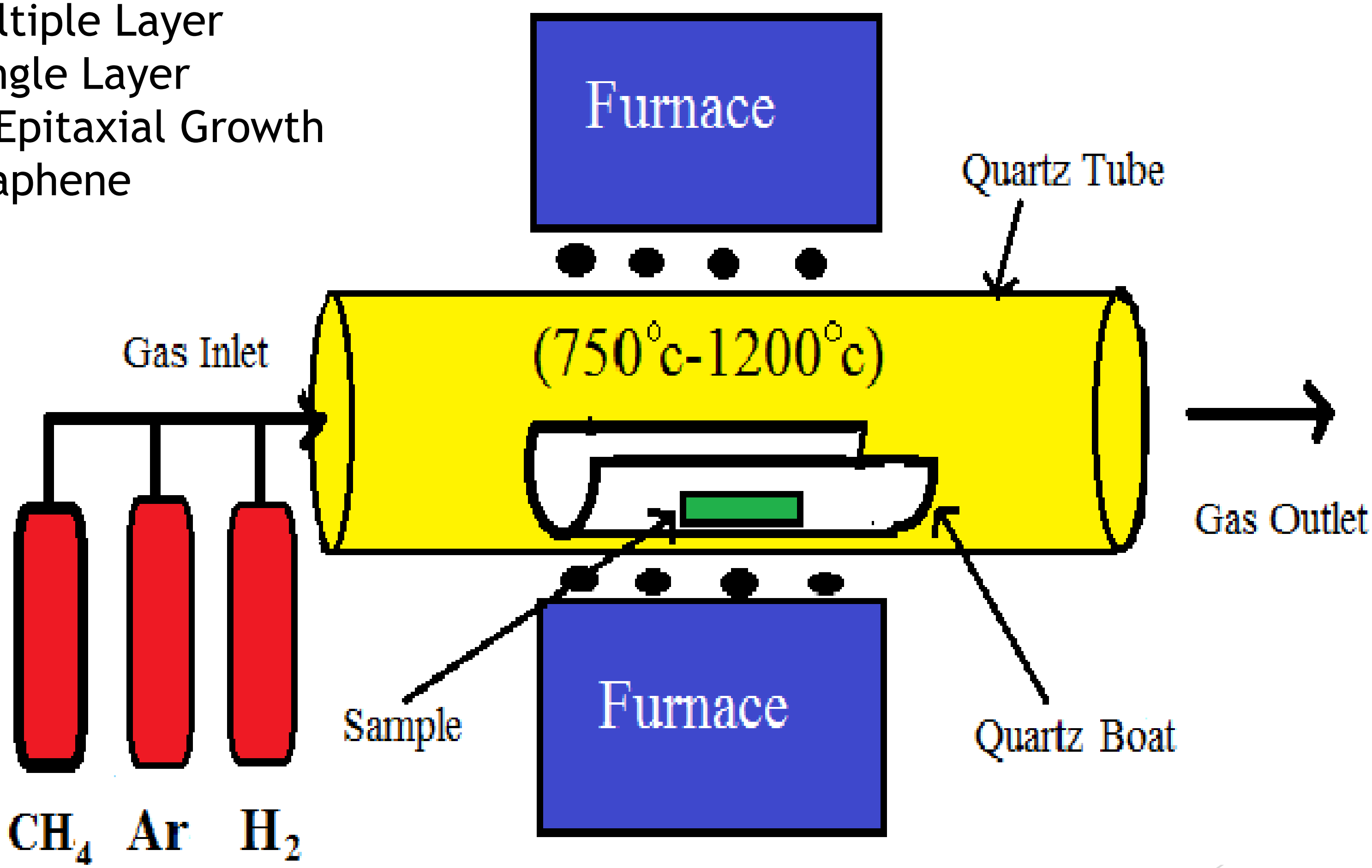
Chemical Exfoliation



Graphene Synthesis

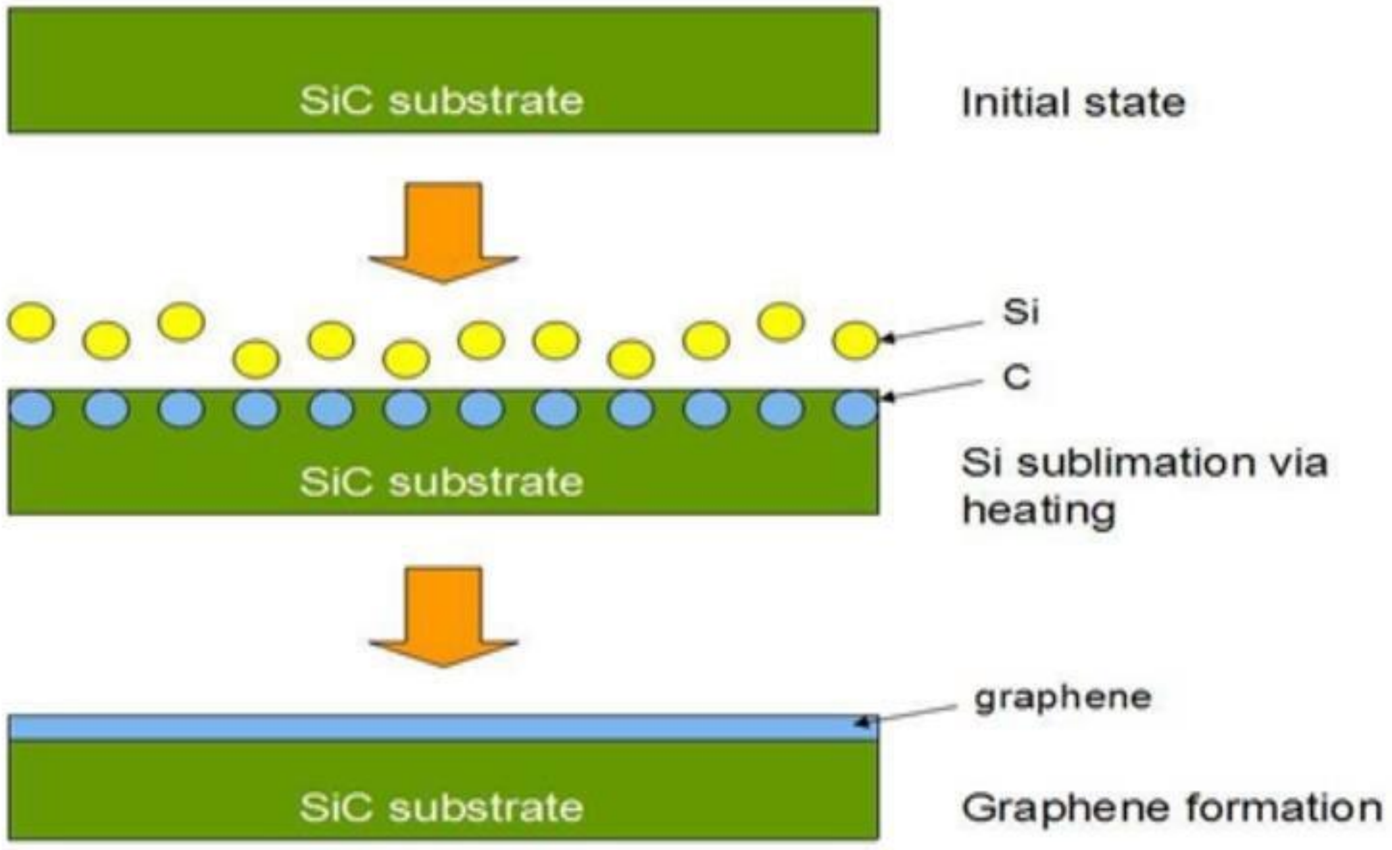
Chemical Vapor Deposition (CVD)

Ni Multiple Layer
Cu Single Layer
h-BN Epitaxial Growth of Graphene



Graphene Synthesis

Epitaxial Growth of Graphene via Chemical Vapor Deposition (CVD)



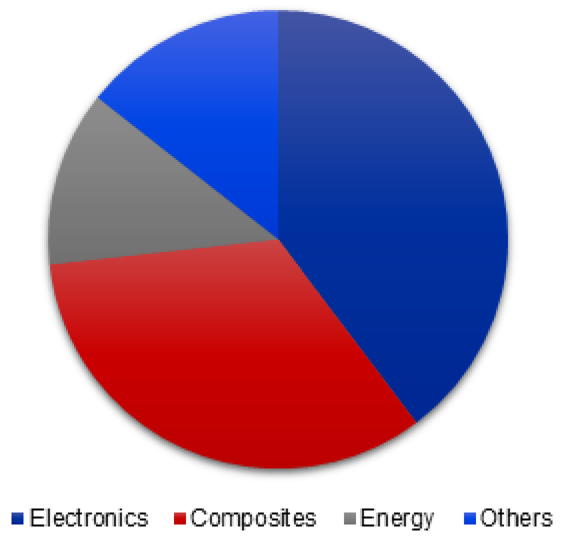
Application of Graphene

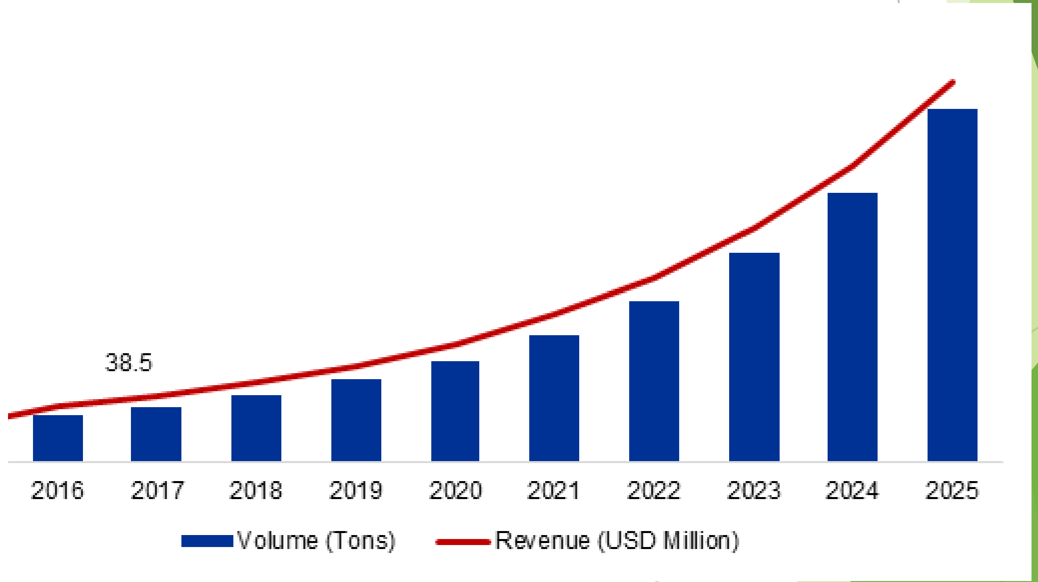
- Energy Storage Application
- Electronics
- Solar Panel

Market Value of Graphene

2017 38 Million

2025 2000 Million





Application of Graphene

Owner EU
Funding: 1 Billion Euro
Objective: Commercialize Graphene Product
Europe's biggest scientific research initiative

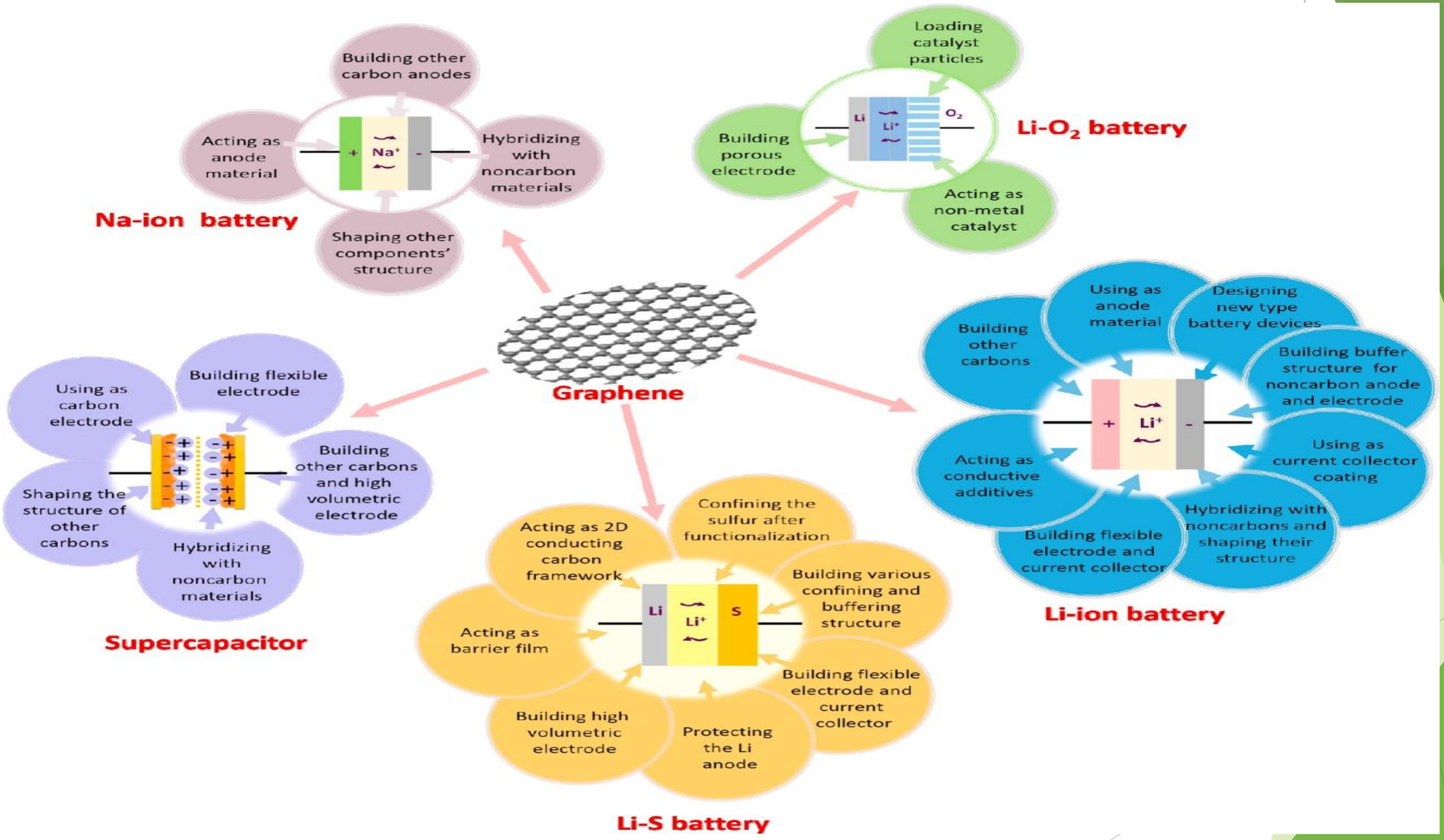
Low-Cost, High-Efficiency III-V Photovoltaics Enabled By Remote Epitaxy through Graphene
Funding: $1,500,000
Objective: develop low-cost, high-efficiency photovoltaics (PV)

The National Graphene Institute,
The University of Manchester

Funding: 61 Million Euro

Energy Storage Application

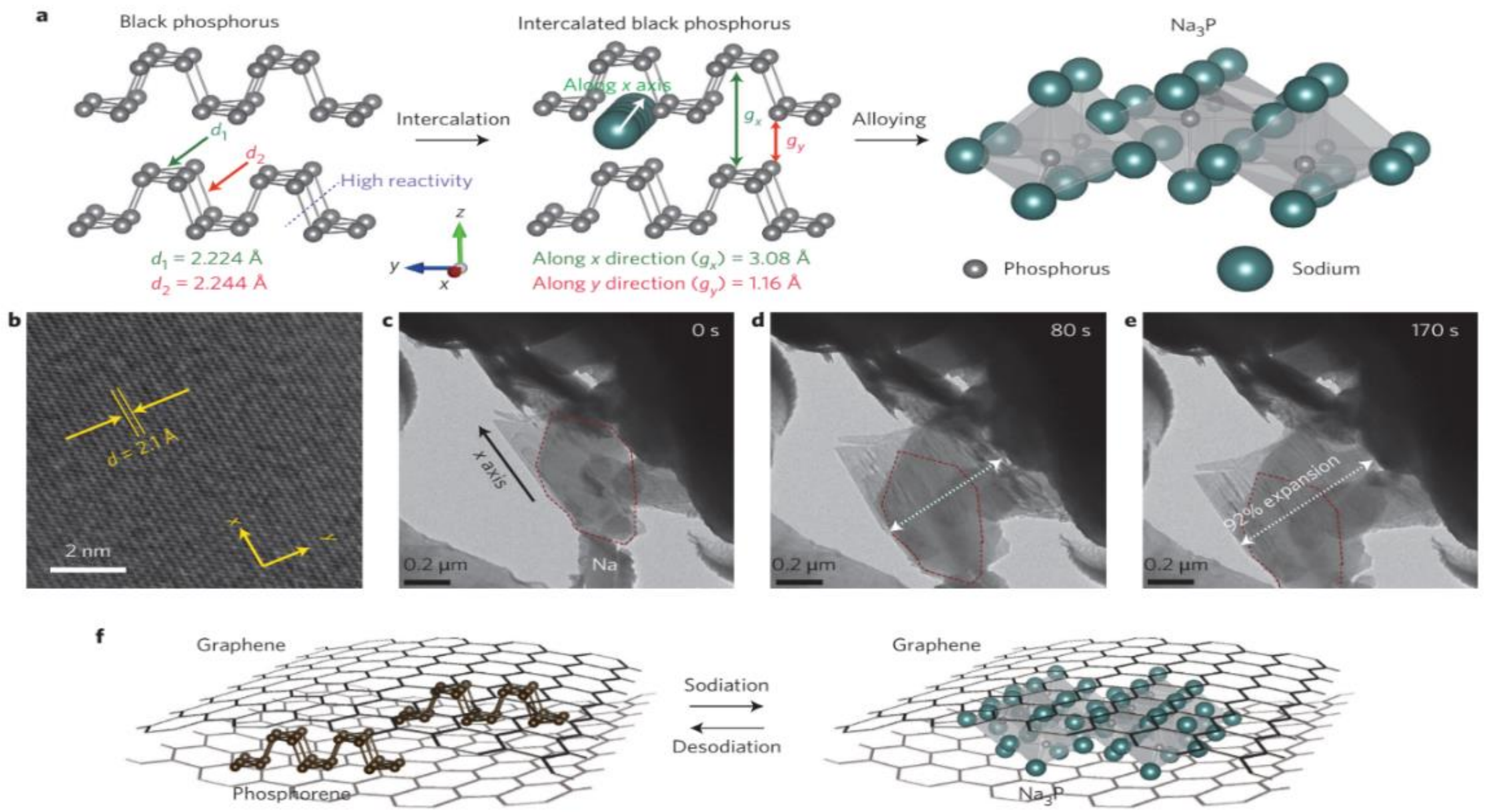
Application of Graphene



Lv, W., Li, Z., Deng, Y., Yang, Q. H. & Kang, F. Graphene-based materials for electrochemical energy storage devices: Opportunities and challenges. *Energy Storage Mater.* **2**, 107–138 (2016).

Application of Graphene

Energy Storage Application

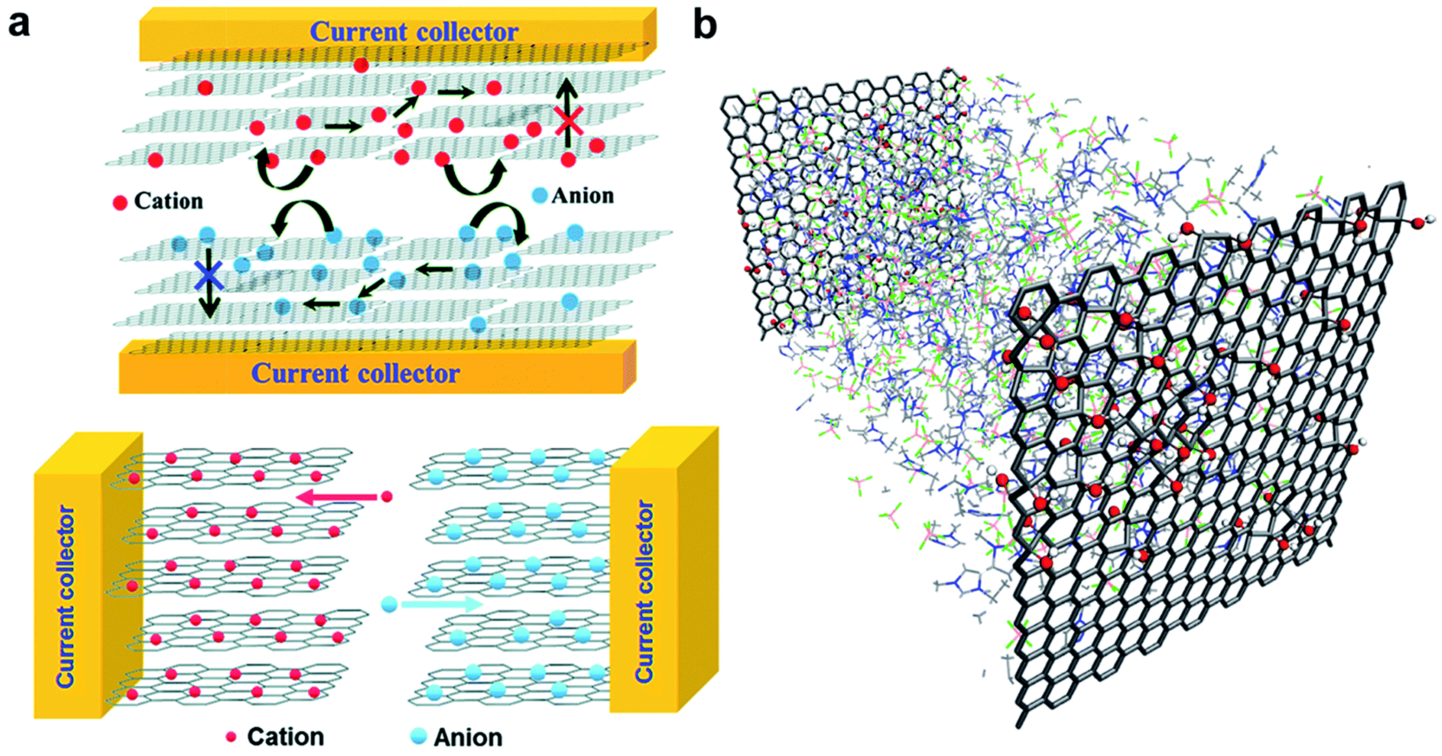


Sodiation in Phosphorene-Graphene Hybrid Material

Sun, J. *et al*. A phosphorene-graphene hybrid material as a high-capacity anode for sodium-ion batteries. *Nat. Nanotechnol*. **10**, 980–985 (2015)

Application of Graphene

Energy Storage Application
Supercapacitors

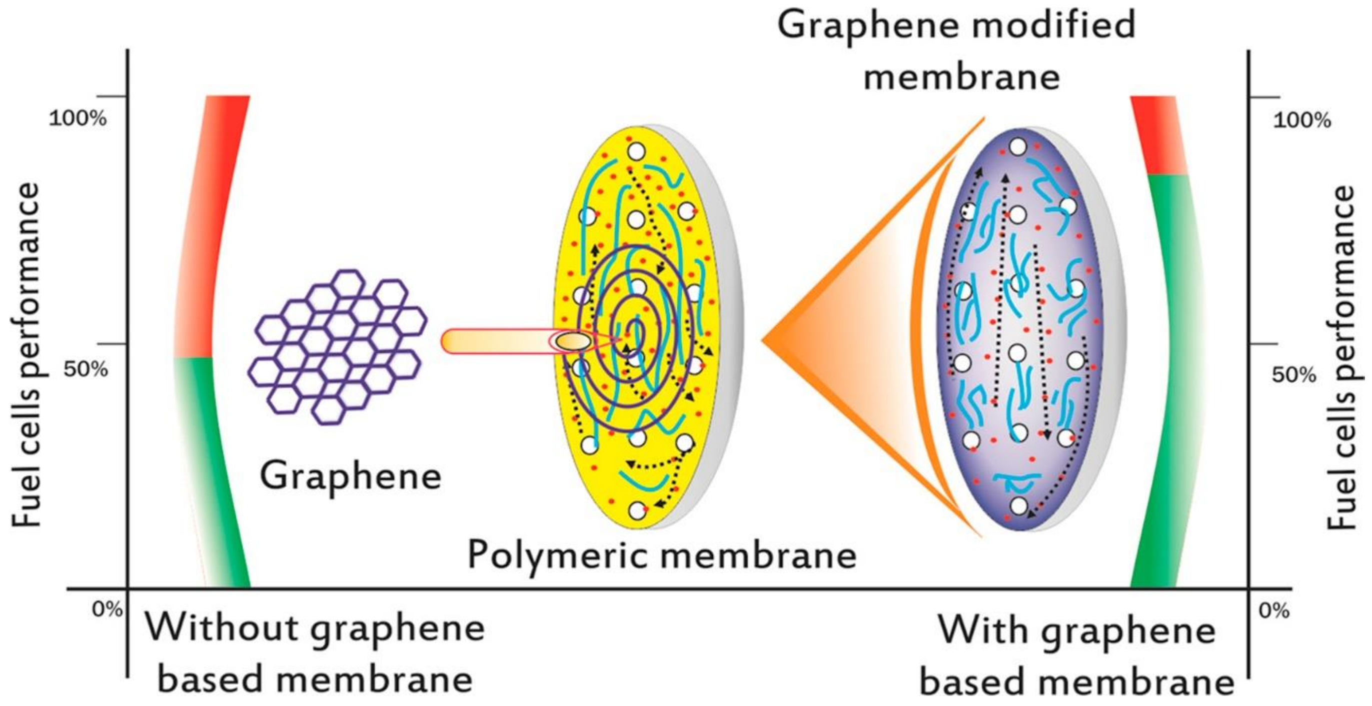


Thomas, J. Environmental. **8**, (2015).

Application of Graphene

Energy Storage Application

Fuel Cells



Farooqui, U. R., Ahmad, A. L. & Hamid, N. A. Graphene oxide : A promising membrane material for fuel c
. *Renew. Sustain. Energy Rev.* **82**, 714–733 (2018)

Conclusion

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